GeoNLI: Natural Language Interpretation of Satellite Imagery

Team 65

***Abstract*—With the rapid expansion of satellite imaging programs, decision-makers in fields such as disaster management, agriculture, defense, and urban planning increasingly require fast and reliable interpretation of geospatial data. Manual analysis or traditional GIS workflows are often too slow to support time-critical scenarios, especially when large regions must be monitored continuously. Therefore, there is an urgent need for advanced systems that can handle the complete spectrum of visual understanding tasks in satellite imagery i.e., from detecting objects and identifying important regions to answering user queries and generating meaningful captions, without requiring expert knowledge. Such automated geospatial perception tools can significantly reduce operational latency, improve situational awareness, and democratize access to Earth observation insights by enabling non-experts to interpret complex imagery with ease. http://167.71.225.78:5173**

I. Introduction

This paper addresses the challenge of interpreting geospatial satellite imagery by introducing a unified framework that integrates visual grounding, captioning, and visual question answering (VQA) capabilities. The approach leverages RL (Reinforcement learning) fine-tuned vision-language models in addition to SOTA Segmentation models to bridge the gap between raw satellite data and human-interpretable insights, providing a comprehensive solution that is both accurate and adaptable across diverse geospatial analysis tasks. Additionally, the framework is fully deployed and operational, demonstrating its practical viability for real-world applications in remote sensing, environmental monitoring, and spatial intelligence. Our contributions are threefold:

1) We propose a unified architecture capable of performing multiple geospatial tasks like Captioning , VQA and Grounding within a single framework, eliminating the need for task-specific models.

2) We curate and utilize a large-scale corpus of over 30,000 annotated data points, encompassing caption-only, VQA-only, and visual grounding samples to ensure comprehensive model training and task generalization.

3) We conduct rigorous evaluation across diverse benchmark datasets, demonstrating that our approach achieves robust performance across different geographical regions, image resolutions, and query complexities.

This work not only advances the technical discourse on vision-language understanding for satellite imagery but also provides a scalable and deployable solution, paving the way for its practical application in critical domains.

II. Targeted Problems

A. Captioning

Image captioning is a fundamental vision-language task that aims to generate natural language descriptions that accurately and coherently describe the visual content of an image. This task requires models to not only recognize objects and their attributes within the scene but also understand spatial relationships, actions, and contextual information to produce human-like textual descriptions. The challenge lies in bridging the semantic gap between visual features and linguistic expressions while maintaining grammatical correctness, semantic accuracy, and descriptive richness. Existing image captioning methods have progressed through several paradigms: early template-based and retrieval-based approaches that either fill predefined sentence templates or retrieve similar captions from databases; encoder-decoder architectures that employ CNNs as visual encoders and RNNs or LSTMs as language decoders with attention mechanisms to focus on relevant image regions during word generation; and more recent transformer-based models which uses VIT as vision backbone like llava , Qwen that leverage self-attention and cross-attention mechanisms for better long-range dependency modeling and cross-modal alignment. Contemporary approaches have also explored reinforcement learning techniques using metrics like CIDEr [1] and SPICE [2] as rewards to optimize caption quality, as well as controllable captioning methods that generate descriptions conditioned on specific attributes, styles, or user preferences, thereby enhancing both the flexibility and applicability of captioning systems.

B. Visual Grounding

Visual grounding is a fundamental vision-language task that aims to localize specific objects or regions in an image based on natural language referring expressions. The task requires models to establish precise correspondences between linguistic descriptions and their corresponding visual entities, effectively bridging the semantic gap between textual and visual modalities. Unlike traditional object detection with predefined categories, visual grounding demands fine-grained understanding of spatial relationships, attributes, and contextual information embedded in free-form natural language queries. Recent advancements leverage large-scale pre-training, broadly diverging into two distinct paradigms. The first category treats grounding as an extension of **multimodal conversational reasoning**, where systems integrate localization directly into dialogue streams. These models process user queries to output spatial coordinates alongside textual responses, effectively merging high-level semantic interpretation with object detection. Conversely, the second category approaches the task through **promptable segmentation**, prioritizing pixel-level precision. These vision-centric foundation models utilize text embeddings as prompts to generate dense, class-agnostic masks, focusing on exact boundary delineation rather than conversational interactivity.

C. Visual Question Answering

Visual Question Answering (VQA) is a challenging multimodal task that requires models to answer natural language questions about visual content by jointly understanding and reasoning over both image and text modalities. VQA encompasses several answer types based on the nature of questions and expected responses: *numeric VQA* involves counting objects or quantifying attributes in images (e.g., "How many cars are in the parking lot?"), requiring models to perform counting and numerical reasoning; *semantic VQA* demands understanding of object attributes, scene context, and semantic relationships to answer questions about colors, actions, and object properties (e.g., "What color is the woman's dress?") and *boolean VQA* requires binary yes/no responses to questions about the presence, state, or relationships of entities in the image (e.g., "Is the dog sitting on the couch?"). Recent state-of-the-art models leverage large-scale pre-trained vision-language models including BLIP [3], BLIP-2 [4], and Flamingo [5], which demonstrate strong performance through unified encoder-decoder architectures and multimodal pre-training strategies. Furthermore, the emergence of large multimodal models such as GPT-4V [6], LLaVA [7], and Qwen-VL [8] has pushed the boundaries of VQA by integrating large language models with visual encoders, enabling more sophisticated reasoning and chain-of-thought explanations for complex visual questions.

III. Existing Approaches

Both multi-task and task-specific vision–language models were initially explored to identify architectures capable of supporting visual grounding, visual question answering, and caption generation in remote sensing. Promising generalization across multiple objectives is demonstrated by multi-task approaches such as GLaMM [9] and GeoPixel [10], while strong performance on their respective tasks is exhibited by task-specialized solutions including SAM-3 [11] for segmentation, GroundingDINO [12] for grounding, and EarthDial [13] for grounding and VQA. It has been suggested by prior research that superior accuracy is often achieved by task-specific models due to architectures and learning objectives optimized for individual capabilities. In the following subsections, the most relevant representative approaches are discussed.

A. EarthDial

EarthDial is a specialized vision–language model designed for Earth Observation data, built upon the InternVL framework with architectural enhancements to support multi-spectral and multi-temporal imagery. The architecture comprises a vision encoder (InternViT), a multilayer perceptron (MLP) projector and a large language model (Phi-3-mini [14]). Two core innovations enable robust handling of remote sensing data. First, the adaptive high-resolution module dynamically selects aspect ratios and generates image patches to preserve fine-grained spatial features across varying resolutions. Second, a data fusion block aggregates spectral and temporal information into unified visual tokens through iterative cross-channel processing. These visual tokens are projected into the language model's token space and combined with user queries.

EarthDial supports multimodal interaction, enabling visual grounding, caption generation and VQA within a unified conversational interface. It can localize objects, respond to region-specific queries and produce natural-language scene interpretations.

Limitations:

- It is incapable of generating accurate floating-point numerical answers in numeric VQA and performs reliably only for integer-based queries.
- It struggles with grounding when a target class has high instance density (more than 5 occurrences), often missing objects or predicting incorrect bounding boxes.
- The model is not instruction-tuned, making it unable to follow complex or multi-step instructions within a single query.

B. SAM-3

SAM-3 (Segment Anything Model 3) is a foundation model for promptable segmentation in both images and videos, introducing Promptable Concept Segmentation (PCS). It is built around a shared vision backbone coupled with two complementary components: a perception encoder (DETR-based object detector) and a memory-based tracker for temporal consistency in videos. This decoupled design prevents cross-task interference while supporting comprehensive segmentation.

SAM-3 accepts multimodal prompts including text descriptions, reference examples or combination inputs. Visual embeddings are aligned with text embeddings through cross-attention to generate instance-level masks and identifiers. The model demonstrates strong few-shot generalization, making it well-suited for domains with limited manual annotation.

Limitations:

- SAM-3 is not a multimodal VLM and cannot perform captioning or VQA tasks.

C. Other Vision–Language Approaches

Apart from the above mentioned models, several recent multimodal frameworks extend grounded reasoning to natural and remote sensing imagery. GLaMM enables visually grounded dialogue, region-level captioning and segmentation by combining a global encoder, region encoder, pixel decoder and large language model. GeoPixel adapts this pixel-grounding paradigm to overhead imagery using an adaptive image divider, CLIP-based encoder and grounding module to support analysis of high-resolution satellite scenes.

However, existing approaches face domain-specific challenges. GLaMM is primarily trained on natural-image datasets and struggles with scale variation, nadir viewpoint and multi-sensor signals inherent to remote sensing. GeoPixel improves pixel-level grounding in overhead scenes, but remains limited to RGB imagery and does not provide a lightweight unified conversational framework covering the broad range of tasks including SAR and multispectral reasoning, scene-level semantics and diverse VQA.

IV. Training Data

To develop a reliable model capable of visual grounding, visual question answering and geospatial captioning it is essential to have a diverse and extensive dataset with a sufficient variety and

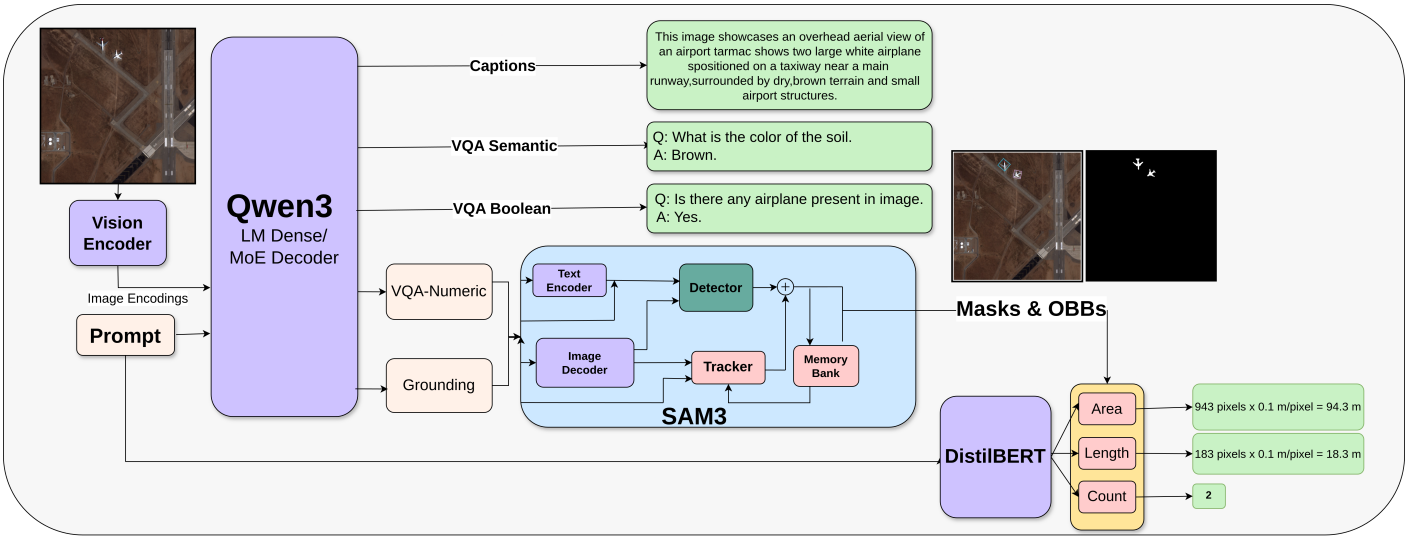


Fig. 1. End-To-End Multi-Modal Reasoning and Segmentation Pipeline

number of samples to ensure robust performance. Consequently, constructing a high-quality dataset for fine tuning our models was a critical initial step.

The fine-tuning dataset is curated from a wide range of publicly available remote sensing datasets. From GAIA, we leverage high-resolution satellite images paired with detailed multi-sentence captions. To increase semantic diversity, we include scene classification and captioning datasets such as RESISC45 and RSICD. For grounding and VQA supervision, datasets from EarthDial and related resources including RSVG, DIOR-RSVG, MQVQA, DOTA and RSIVQA are incorporated.

Additionally, multispectral imagery from NASA's HLS S30 and SAR scenes sourced through the Alaska Satellite Facility interface are included to broaden spectral and temporal coverage. Our approach involves taking SAR (VV, VH, VV/VH) and IR (Red, Green, NIR, SWIR) bands to create False Color Composites (FCCs) that emphasize aspects such as vegetation, soil, water bodies, or urban areas. We leverage an existing close sourced LLM with specific prompts and FCC to generate annotations due to unavailability of public annotated SAR/IR datasets . All datasets are co-registered, clipped to consistent spatial extents and filtered to remove corrupted samples. The complete corpus is harmonized into a unified structure with fields for imagery, spatial annotations and natural language supervision.

TABLE I
DATASET SUMMARY

Dataset Type	Source
VQA	MCVQA
VQA	Sydney_VQA
VQA	UCM_VQA
Grounding	DIOR-RSVG
Grounding	RSVG
Grounding	DOTA
Captioning	GAIA
Captioning	RESISC
Captioning	RSICD
SAR	NASA-Earthdata
IR	NASA-Earthdata

TABLE II
DISTRIBUTION OF IMAGE RESOLUTIONS IN TRAINING SET

Resolution	Count	Fraction
256 × 256	12,400	42.0%
800 × 800	8,200	28.0%
1024 × 1024	3,500	12.0%
224 × 224	2,900	10.0%
2048 × 2048	1,700	6.0%
600 × 600	600	2.0%

V. PROPOSED FRAMEWORK

We present a unified framework designed for comprehensive visual understanding, capable of bridging the gap between high-level semantic reasoning and low-level pixel-wise segmentation. The pipeline, illustrated in Figure 1, operates in three stages:

1) **Semantic Reasoning:** Qwen3-VL-8B interprets user prompts and visual inputs to generate semantic answers and captions.
2) **Segmentation based Grounding:** SAM-3 utilizes the textual query and visual embeddings to segment out the required objects and areas.
3) **Quantitative Attributes Computation :** Further these segments are used to compute the required quantities like Area , Length , Perimeter & Count .

A. Visual-Language Reasoning (Qwen3-VL)

The backbone of our system is the Qwen3-VL architecture, which serves as the primary reasoning engine.

a) Input Processing: Let $I \in \mathbb{R}^{H \times W \times 3}$ denote the input image, and T_{prompt} denote the user's natural language query (e.g., "Describe the image"). The image is processed by a Vision Encoder (i.e., SigLip2-So400M) to yield a sequence of visual tokens X_v. These are concatenated with the tokenized text prompt X_t, and then passed to Decoder.

b) Dense/MoE Decoding: The unified sequence $[X_v; X_t]$ is fed into the Qwen3 LM Decoder. This decoder, configurable as either a Dense or Mixture-of-Experts (MoE) architecture, models the conditional probability of the output sequence Y:

$$P(Y \mid I, T_{prompt}) = \prod_{i=1}^{L} P(y_i \mid y_{<i}, X_v, X_t) \tag{1}$$

c) Task-Specific Heads: The model is fine-tuned to produce three distinct categories of outputs based on the prompt type:

- **VQA-Semantic:** Generates natural language answers to semantic questions (e.g., "The color of the soil is brown.").
- **VQA-Boolean:** Answers a query as a binary(yes/no) judgment.
- **Captioning:** Generates a highly descriptive and semantically rich caption for the image.

B. Unified Segmentation & Grounding (SAM 3)

SAM3 is employed to generate semantic and instance segmentation mask conditioned on user query . This module integrates an open-vocabulary **Detector** with a temporal **Tracker**.

1) Input Configuration: SAM 3 handles two types of tasks: Promptable Concept Segmentation (PCS) (i.e. prompt interpretation with respect to visual features) and Point Visual Sampling (PVS) (i.e. selection of high-fidelity patch for further operations).

- **Spatial Prompts:** Bounding boxes and points as visual prompts are used to segment the objects.
- **Semantic Prompts:** Abstract queries are generated by the Qwen3 VL as the model tends to perform better for short and specific queries . Then the generated query is passed as a text prompt to SAM for text-conditioned segmentation.

We have utilized only Semantic prompts in our pipeline

2) Parallel Processing Streams: The module operates through Detector Stream. It employs a cross-modal attention mechanism where text embeddings E_{txt} act as queries against image features F_{img}:

$$M_{\text{det}} = \text{Decoder}\left(\text{CrossAttn}(E_{\text{txt}}, F_{\text{img}}), B_{\text{pred}}\right) \tag{2}$$

This generates masks M_{det} aligned with both the semantic description and spatial constraints.

3) Fusion: The outputs from the stream produces the set of binary masks $\mathbf{M}_{\text{final}}$ and Horizontal Bounding Boxes (HBBs). For getting appropriate instance masks, we had two methods. Directly picking instance masks above a certain threshold, we found this method to be too much dependent on the threshold and hard to find an optimal threshold. What we instead did is, obtain instance masks with score ≥ 0.1 (The reason for picking 0.1 is to filter out the completely random boxes)

Now, for each Instance Bounding Box(IBB), we compute it's intersection area with semantic mask, divided by area of IBB, and filter the IBBs with this value $\geq threshold(t2)$. But this doesn't filter cases where same objects might end up with multiple boxes, so we further apply Non-Max Suppression [15](NMS), to remove duplicate or highly overlapping boxes.

C. Quantitative Attributes Computation

The final stage extracts actionable metric data from the geometric outputs of SAM 3. The Distill-Bert [16] model acts as a function caller for the geometric operations. Given the set of masks $\mathbf{M}_{final}$ and the query type, the module computes:

1) Area: The area and perimeter/length functions take the binary segmentation semantic mask produced by SAM 3 and compute its area and perimeter using contour geometry. First, each function extracts the external contours from the mask using cv2.findContours, then selects the largest contour, assuming it represents the main object. The area function uses cv2.contourArea to compute the number of pixels inside this contour; if a physical resolution (meters per pixel) is provided, it converts pixel-area into real-world area, otherwise it simply returns the pixel count.

2) Perimeter/Length: The perimeter function uses cv2.arcLength to measure the length of the contour; again, if a spatial resolution is provided, the value is converted into real-world units.

3) Count: The count of the components in Instance Mask generated after applying NMS.

VI. Fine-Tuning Methodologies

A. Qwen-3-VL-8B RL Finetuning

For inducing query conditioned detailed reasoning over images in the vlm we employed RL based fine tuning methods. One of the most standard reinforcement learning fine-tuning techniques is **Proximal Policy Optimization (PPO) [17]**. The process was laid out in the InstructGPT paper to create a model that can follow instructions and go beyond simple next-word prediction. However, the fine-tuning process requires substantial labeled data along with expert-ranked responses, which are difficult to collect. Additionally, PPO is computationally expensive to train because it employs four models during policy optimization: the *Policy*, *Reference*, *Reward*, and *Value* models. Both the policy and value models contain trainable parameters that require backpropagation, demanding significant memory and computational resources. Furthermore, PPO requires a critic model to calculate advantages, and training a stable critic throughout this optimization process is challenging due to the non-stationary distribution of states and actions generated by the policy, which can sometimes lead to divergence from inaccurate critic responses.

The DeepSeek-R1 paper introduced **Group Relative Policy Optimization (GRPO)** [18], which offers a more efficient and stable approach to training reasoning models by removing the value model, thereby reducing computational requirements and memory usage. The value model estimates the quality of generated tokens; to eliminate its need, GRPO generates multiple outputs per user query. For each output, it calculates a reward using multiple reward functions, and the advantage is computed by normalizing over the group rewards as illustrated in Fig. 2. The final component in this optimization process is the KL divergence term, calculated using the reference model (a frozen copy of the original model) to prevent the policy model from drifting too far from its initial state. This also prevents reward hacking behaviors, such as the model learning to output specific words like "addCriterion" or excessive newlines to achieve higher rewards. Reward hacking can be further mitigated by introducing specific reward functions.

$$r_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t}|q,o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q,o_{i,<t})}, \quad A_i = \frac{r_i - \text{mean}(r_1,\ldots,r_G)}{\text{std}(r_1,\ldots,r_G)} \tag{3}$$

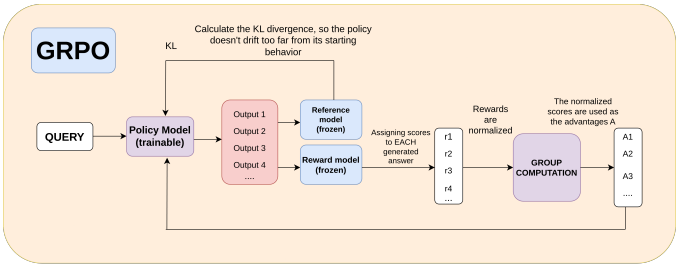


Fig. 2. GRPO Framework

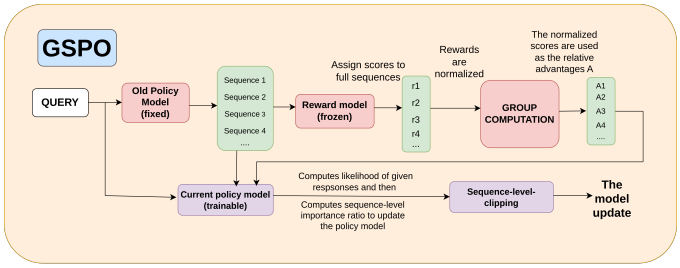


Fig. 3. GSPO Framework

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\min(r_{i,t}(\theta)A_i, \text{clip}(r_{i,t}(\theta), 1-\epsilon, 1+\epsilon)A_i - \beta D_{\text{KL}}(\pi_\theta \,\|\, \pi_{\text{ref}})\right] \tag{4}$$

The reward functions employed in our VLM fine-tuning process include:

- **Formatting Reward Function:** Rewards the model for generating query-specific reasoning over images within `<REASONING>` and `</REASONING>` tags.
- **Correctness Reward Function:** Calculates the BLEU score between the answer enclosed in `<SOLUTION>` and `</SOLUTION>` tags and the ground truth answer.

However, GRPO's objective is ill-posed due to a mismatch between reward granularity (sequence-level) and optimization granularity (token-level). Many tokens get clipped, leading to inefficient learning. This problem becomes particularly acute when training large models on long-response tasks, potentially resulting in catastrophic model collapse. Since our aim is to train a VLM for detailed reasoning over images, we must forego the token-level objective and perform optimization directly at the sequence level. Therefore, we adopt a more stable and faster optimization process using **Group Sequence Policy Optimization (GSPO) [19]**, introduced by the Qwen team, where all tokens share the same length-normalized sequence-level importance ratio, reducing per-token noise as illustrated in Fig 3

$$s_i(\theta) = \exp\left(\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\log\frac{\pi_\theta(o_{i,t}|q,o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q,o_{i,<t})}\right) \tag{5}$$

B. SAM3 Custom loss Fine-Tuning

While the Segment Anything Model 3 (SAM3) exhibits strong zero-shot capabilities, its direct application to remote sensing imagery is hindered by domain-specific challenges. Satellite and aerial imagery frequently contain tiny, densely packed objects and low-contrast regions that standard pre-trained weights fail to segment accurately. To address this, we implement a fine-tuning procedure tailored to multi-resolution images and diverse user queries.

To adapt SAM3 computationally efficiently, we employ Low-Rank Adaptation [20] (LoRA). Instead of updating the full weight parameters, we inject trainable rank decomposition matrices into the Transformer architecture. Specifically, we target the query (W_q) and value (W_v) matrices of the attention modules with a rank $r = 8$. For a pre-trained weight matrix $W_0 \in \mathbb{R}^{d\times k}$, the update is given by:

$$W' = W_0 + BA \tag{6}$$

where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ are trainable parameters, effectively adapting the model to the remote sensing domain while keeping the vast majority of parameters frozen.

To maximize the Mean Intersection over Union (mIoU) and ensure robust mask selection, we employ a hybrid objective function combining supervised segmentation loss with a custom Reinforcement Learning (RL) based alignment loss.

1) Supervised Loss ($\mathcal{L}_{sup}$): The foundation of the training utilizes a combination of Dice Loss and Binary Cross Entropy (BCE) Loss. This encourages pixel-level accuracy and penalizes local segmentation errors.

$$\mathcal{L}_{sup} = \lambda_{dice}\mathcal{L}_{Dice} + \lambda_{bce}\mathcal{L}_{BCE} \tag{7}$$

However, supervised loss alone is insufficient. It primarily penalizes local pixel discrepancies but fails to capture the global interaction between false positives and false negatives. Furthermore, mIoU is a non-differentiable, non-local metric; optimizing pixel-level entropy does not guarantee optimal mIoU scores during evaluation.

2) GRPO-Inspired IoU Preference Loss ($\mathcal{L}_{RL}$): SAM3 outputs a set of candidate masks $M = \{m_1, m_2, \ldots, m_k\}$ and a corresponding probability score distribution $P(m)$. A critical limitation of standard training is that a candidate mask with a higher ground-truth IoU may be assigned a lower probability score by the model, as supervised loss typically optimizes only the "best" mask index.

To align the model's confidence with actual mask quality, we introduce a Group Relative Policy Optimization (GRPO) inspired loss. This formulation treats mask generation as a policy and introduces stochasticity to encourage exploration of the mask space.

For a given query, we sample a group of G candidate masks based on the model's predicted distribution. For each mask m_i, we calculate the reward R_i as the IoU with the ground truth mask M_{gt}:

$$\text{IoU} = \frac{|\hat{m} \cap m|}{|\hat{m} \cup m|}$$

$$R_i = \text{IoU}(m_i, M_{gt}) \tag{8}$$

TABLE III
CAPTIONING AND VQA RESULTS ON *VRSBench*.

Model	Captioning (BERT-BLEU4)	Boolean QA	Numeric QA	Semantic QA
Earthdial	-	0.96	0.56	-
Earthdial*	0.84	0.89	0.53	0.72
Llava 1.5***	0.9737	0.52	0.35	0.68
Qwen3 VL RL tuned*	0.9888	0.7676	0.557	0.7443
Qwen3 VL RL tuned***	-	0.81	0.36	0.92

$$L_{RL} = -R \cdot \log \pi_\theta(\hat{m})$$

The RL objective is to maximize the probability of masks that result in a positive advantage (higher IoU than the group average) while suppressing those with negative advantage.
Both Qwen3VL and SAM3 were fine tuned on 80gb Vram A100 GPU.

VII. EVALUATION

A. Captioning

Captioning is evaluated using **BERT-BLEU [21]**, which combines BLEU-n precision with semantic similarity from BERT embeddings, capturing both lexical and contextual alignment.

Let candidate tokens be $C = \{c_1, \ldots, c_m\}$ and reference tokens $R = \{r_1, \ldots, r_k\}$, with maximum n-gram order $N = 4$. Let $E(x)$ denote the BERT embedding of x, and similarity be $\cos(E(a), E(b))$.

1) Semantic n-gram Precision: n-gram sets:

$$C_n = \{c_{i:i+n-1}\}, \qquad R_n = \{r_{j:j+n-1}\}$$

Semantic precision:

$$P_n = \frac{1}{|R_n|} \sum_{r \in R_n} \max_{c \in C_n} \cos(E(c), E(r))$$

2) BERT-BLEU Score:

$$\text{BERT-BLEU}_N = LP \cdot \left(\max_n P_n \right)$$

Length penalty:

$$LP = \exp\left(-\alpha \cdot \frac{|L_C - L_R|}{L_R} \right), \qquad \alpha = 0.5$$

Higher scores indicate stronger semantic and lexical similarity between generated and reference captions.

B. Grounding

Grounding evaluates how well a model links textual references to visual regions by checking (1) the number of detected instances and (2) the spatial accuracy of their bounding boxes. The evaluation metric combines a count penalty with the mean Intersection over Union (MeanIoU).

$$S_{\text{grounding}} = CP \times \text{MeanIoU}$$

$$\text{CP} = \exp\left(-\alpha \left| N_{\text{pred}} - N_{\text{ref}} \right|\right), \alpha = 23$$

where CP is Count Penalty, N_{pred} is the number of predicted instances and N_{ref} the ground-truth count. Higher $S_{\text{grounding}}$ implies better alignment between predicted and reference objects.

TABLE IV
GROUNDING EVALUATION ON *VRSBench*.

Model	Grounding (mIOU)
GeoPixel	0.21
Moondream3	0.29
SAM3	0.33
SAM3***	0.61
EarthDial*	0.38

** indicates fine-tuning on VRSBench trainset.*
*** indicates fine-tuning on RefSegRS and VRSBench.*
**** indicates fine-tuning on the dataset curated by us.*

C. Visual Question Answering

The Visual Question Answering (VQA) evaluation is categorized into three attribute types, each with specific metrics and weightages.

1. Binary Attribute: Evaluates discrete "Yes/No" decisions using an exact match criterion.

$$S_{\text{binary}} = \begin{cases} 1, & \text{if } R_{\text{pred}} = R_{\text{gt}} \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

2. Numeric Attribute : Measures precision using a scale-independent relative error with exponential decay.

3. Semantic Attribute: Assesses contextual meaning using the BERT-BLEU4 metric, combining semantic embedding recall (P_n) with a length penalty (LP).

$$S_{\text{semantic}} = \text{LP} \cdot \max_{n=1}^{N}(P_n) \tag{10}$$

The length penalty is defined as:

$$\text{LP} = \exp\left(-\alpha \cdot \frac{|L_C - L_R|}{L_R} \right), \alpha = 0.5 \tag{11}$$

VIII. SYSTEM ARCHITECTURE AND IMPLEMENTATION

The proposed architecture is designed as a modular, scalable, and privacy-centric application capable of handling complex multi-modal tasks. The architecture decouples the frontend user interface, the application logic (middleware), data persistence, and the high-performance inference servers.

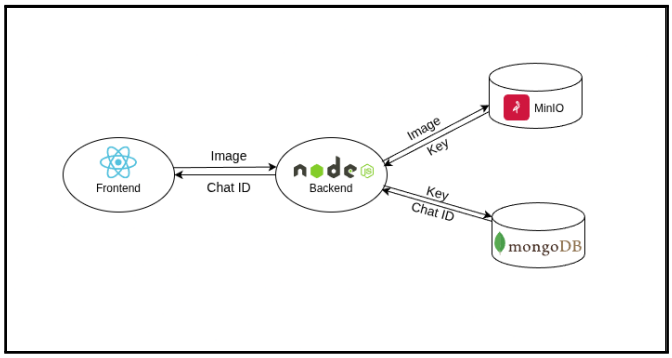


Fig. 4. Image upload and Chat Creation

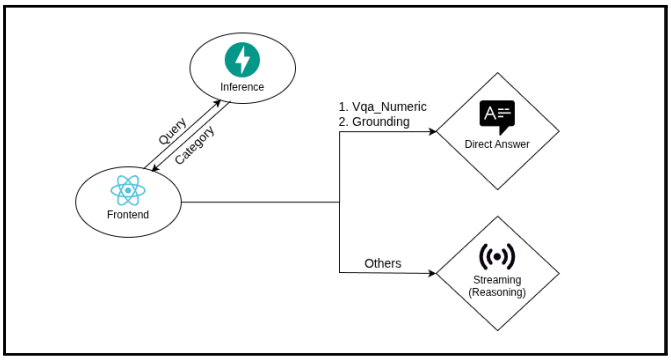


Fig. 5. Automatic Query Classification

A. Workflow and Data Management

The interaction pipeline initiates when a user uploads an image. The application supports both authenticated and guest access, though historical chat retrieval is restricted to authenticated sessions. Upon image upload, the raw binary data (BLOB) is transmitted to the backend, which persists the file in MinIO, an S3-compatible high-performance object storage system. Simultaneously, a chat session entry is initialized in the MongoDB database, and a signed, on-the-fly URL is generated for frontend rendering and downstream inference consumption as illustarted in Fig 4.

To ensure low latency and high availability, the backend (built on Node.js) acts as an asynchronous middleware. It orchestrates communication between the client and the specialized inference services, utilizing threading and request queuing on the inference side to manage load and minimize downtime.

B. Multi-Modal Inference Pipeline

The core reasoning engine employs a dynamic routing mechanism based on query intent. When a user submits a query, it is first processed by a fine-tuned Qwen model hosted on a FastAPI inference server to classify the task into one of five categories: *captioning*, *grounding*, *vqa_semantic*, *vqa_numeric*, or *vqa_binary*.

1) Generative Tasks (Qwen): For queries classified as *captioning*, *vqa_semantic*, or *vqa_binary*, the system establishes a long-lived connection using Server-Sent Events (SSE). The backend bridges the frontend and the Qwen inference server, routing streaming tokens in real-time. This allows the user to view the model's reasoning process and solution incrementally. Parallelly, the complete response is aggregated and stored in the database for persistence as illustrated in 5.

2) Discriminative and Geometric Tasks (SAM3 & BERT): Queries classified as *grounding* or *vqa_numeric* are routed to a fine-tuned SAM3 (Segment Anything Model 3) inference server. Unlike the generative pipeline, this process returns the complete response atomically.

- **Numeric Reasoning:** For *vqa_numeric* tasks, a BERT model is utilized to further sub-classify the intent into *area*, *length*, or *count*, facilitating the extraction of precise geometric metrics.
- **Visual Grounding:** The system generates bounding boxes and segmentation masks, which are returned to the frontend for visualization.

C. Air-Gapped Deployment and Security

A distinguishing feature of this architecture is its capability to operate in fully air-gapped environments, ensuring data sovereignty and eliminating external dependencies.

- **Infrastructure Autonomy:** The system eschews public cloud dependencies (e.g., AWS S3, Firebase) in favor of self-hosted open-source alternatives like MinIO and MongoDB. This ensures no vendor lock-in and allows operation within private, isolated networks.
- **Authentication:** To bypass reliance on external identity providers (e.g., Google OAuth) that require internet connectivity, we implemented a self-contained Role-Based Access Control (RBAC) system. An administrator manages user whitelisting and role assignment, making the system suitable for secure facilities such as research centers.
- **Containerization:** The entire application ecosystem is dockerized. This ensures reproducible environments and simplifies deployment via orchestration scripts, resolving dependency conflicts common in complex ML deployments.

D. User Experience and Interface

The frontend, developed in ReactJS, prioritizes usability and interactivity. Key features include:

- **Interactive Visualization:** Users can click on chat items to render bounding boxes directly on the source image.
- **Session Management:** A draggable sidebar provides access to historical sessions, populated with auto-generated descriptions for rapid identification.
- **Collaboration:** A granular permission management system allows users to share chat contexts securely.
- **Accessibility:** The interface supports dynamic theming (Dark/Light modes) and allows the export of annotated images with bounding boxes.

E. Technology Stack

The technology stack, selected to optimize for performance, scalability, and ease of development, is detailed in Fig. 6 and Table V.

IX. Conclusion

This study proposed a unified framework for the comprehensive interpretation of geospatial satellite imagery by leveraging a multi-stage architecture integrating Qwen3-VL and SAM-3. The

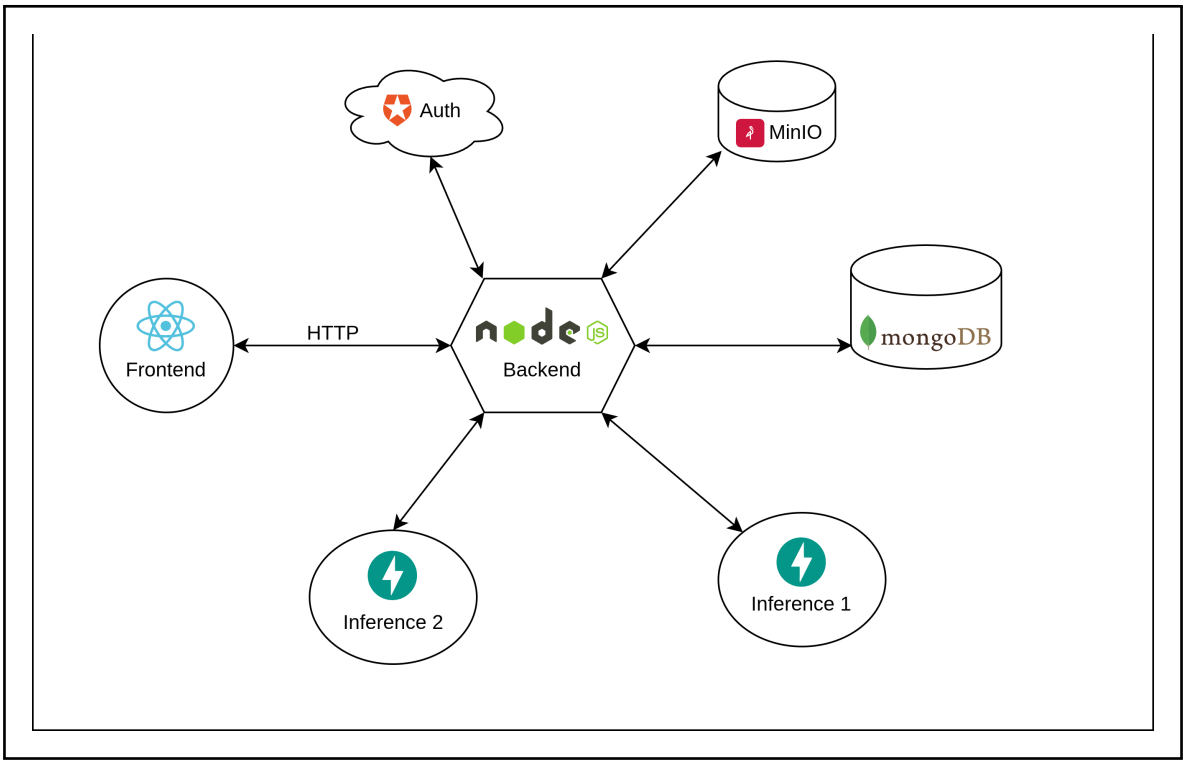


Fig. 6. System Architecture

TABLE V
SYSTEM TECHNOLOGY STACK

Component	Technology & Justification
Frontend	**ReactJS**: Facilitates rapid component-based development and ensures a responsive user interface.
Backend	**Node.js**: Chosen for its event-driven architecture, serving as an efficient middleware for I/O-heavy operations between storage and inference services.
Database	**MongoDB**: A NoSQL document store selected for its flexibility in handling unstructured chat logs and hierarchical query responses.
Storage	**MinIO**: An S3-compatible object storage solution that provides scalability and air-gapped compatibility.
Inference	**FastAPI & Uvicorn**: Provides high-performance, concurrent request handling for model inference, essential for minimizing user wait times.
DevOps	**Docker**: Ensures containerized isolation and streamlined deployment across diverse hardware environments.

approach effectively utilized Reinforcement Learning fine-tuned Vision-Language Models alongside specialized segmentation modules to bridge the gap between raw remote sensing data and high-level human-interpretable insights. By combining visual grounding, captioning, and Visual Question Answering (VQA) capabilities, the framework demonstrated the ability to execute diverse perceptual tasks within a single cohesive system, eliminating the dependency on fragmented, task-specific models.

Our experiments revealed that by employing Group Sequence Policy Optimization (GSPO) for the reasoning engine and a novel GRPO-inspired consistency loss for segmentation, we significantly enhanced the model's alignment between textual queries and pixel-level masks. The integration of a quantitative distillation module further empowered the system to extract actionable physical metrics such as area and length addressing the specific inability of prior works like EarthDial to handle floating-point numeric VQA. Evaluated on the VRS Bench, our approach achieved robust performance, outperforming baselines in captioning (BERT-BLEU4) and semantic reasoning, while proving its practical viability for real-world deployment.

While the results are promising, challenges regarding the scarcity of annotated multi-spectral data remain, particularly for SAR and IR modalities. Future work should focus on expanding the curated corpus beyond the current 30,000 samples and refining the KAN-based autoencoder [22] to further improve the interpretability of latent feature representations. This work represents a key step toward democratizing access to Earth observation insights, paving the way for scalable and automated tools that support time-critical decisions in disaster management and urban planning without requiring expert intervention.

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APPENDIX

A. Integration of SAR and IR Imagery

The biggest bottleneck for SAR and IR VLMs is the lack of high-quality human-written annotations.

1) Proposed Approach: Our approach involves taking SAR (VV, VH, VV/VH) and IR (Red, Green, NIR, SWIR) bands to create False Color Composites (FCCs) that emphasize aspects such as vegetation, soil, water bodies, or urban areas. We leverage an existing LLM (Gemini 2.5 Flash) with specific prompts to generate annotations.

2) Dataset Specifications: The IR dataset utilizes Harmonized Landsat-Sentinel (HLS) data. The SAR dataset is sourced from the Alaska Satellite Facility (ASF) Sentinel-1 mission (GRD-HD type).

B. Feature Engineering and Selection

TABLE VI
FEATURE ENGINEERING SPECIFICATIONS

Modality	Feature Category	Channels / Details
Infrared (IR)	Raw Reflectance	Green, Red, NIR, SWIR1, SWIR2
	Spectral Indices	NDVI (Veg.), NDBI (Urban), MNDWI (Water), NDMI (Moisture)
	Texture Features[a]	13 channels (Gabor, LBP, LPQ, Laws) applied to NIR band
SAR	Basic Intensity[b]	VV, VH, Ratio (VV/VH)
	Polarimetric Arithmetic	Span (Total Power), RVI (Radar Vegetation Index), NDPI
	Texture Features	Gabor Var., LBP, LPQ, Laws Primitives applied to VV & VH

Note: Total feature count: 22 IR channels + 18 SAR channels.
[a] NIR was selected for texture extraction due to maximum contrast between target classes.
[b] An Enhanced Lee Filter is applied to remove speckle while preserving point targets.

C. KAN Autoencoder Engine

The dimensionality reduction engine is built upon a Kolmogorov-Arnold Network (KAN) Autoencoder as illustrated in Fig. 8. Unlike Multi-Layer Perceptrons (MLPs) which use fixed activation functions, KAN places learnable B-Spline functions on the edges connecting neurons. To speed up convergence, the layer includes a standard linear path defined as the base function:

$$y_{\text{base}} = \text{SiLU}(x) \cdot W_{\text{base}} \tag{12}$$

This allows the model to learn simple linear correlations without fitting complex splines. It solves critical problems: learning linear relationships efficiently and preventing dead neurons.

D. Results and Interpretation

1) Latent Feature Analysis: As visualized in Fig. 7, the **X-axis** represents the Normalized Input (Span), while the **Y-axis** denotes the activation strength (coefficients of spline functions). The behavior of each latent dimension is analyzed below:

TABLE VII
INTERPRETATION OF LATENT CURVES

Latent Feature	Observed Behavior	Physical Interpretation
Latent 0 (Blue)	Mostly Flat	Independent of Span; likely represents background noise or bare soil.
Latent 1 (Orange)	Moderate Non-Linearity	Discriminates water bodies (negative activation) from vegetation (positive activation).
Latent 2 (Green)	Weak Non-Linearity	Corresponds to Water (low), Soil (mid), and Urban (high). Characterizes general surface roughness.
Latent 3 (Red)	Strong "Bump"	Shows a distinct peak at intermediate span values, strongly indicating vegetation presence.

E. Advantages of KAN

1) Interpretability: Unlike black-box MLPs, every latent dimension in KAN learns an explicit, visible function of the input.

2) Functional Nonlinearity: The model captures complex nonlinear relationships without requiring deep, stacked layers.

3) Stability: The use of B-splines enforces smooth curves, resulting in disentangled and physically meaningful latent features.

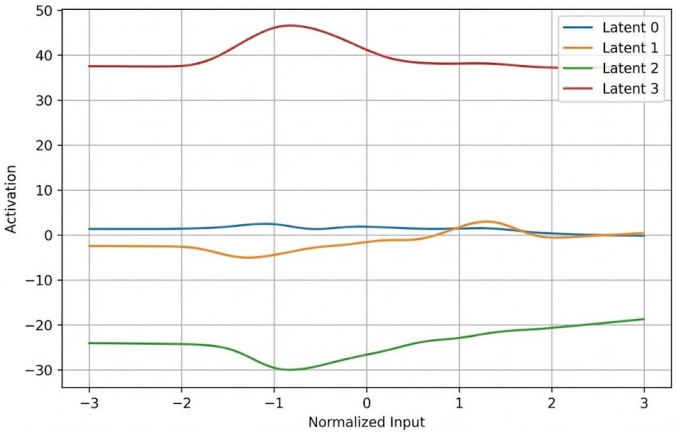


Fig. 7. Interpretation of compressed features derived from span

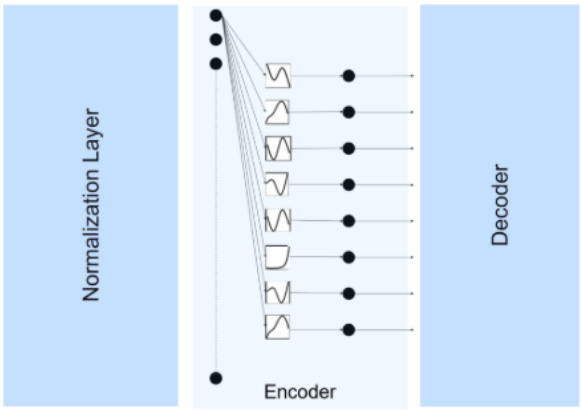


Fig. 8. Feature Compression

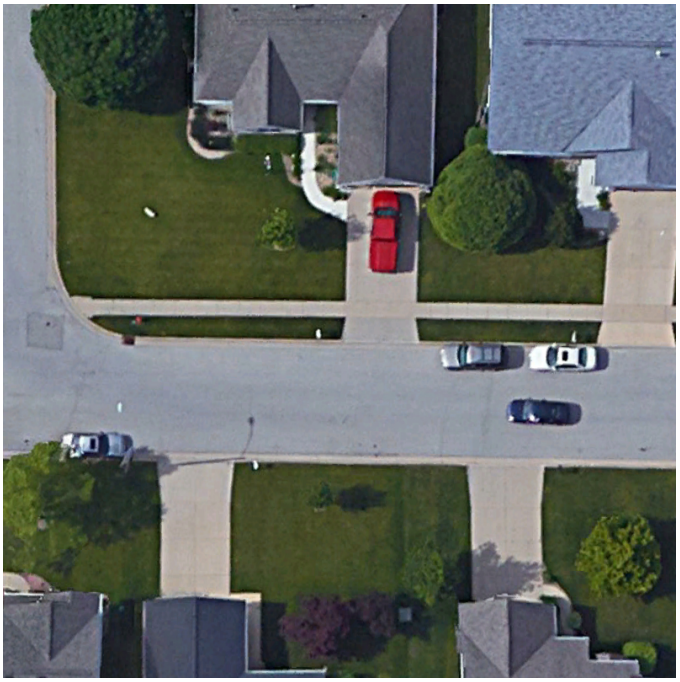


Fig. 9. Baseline SAM3 *prompt:antenna*

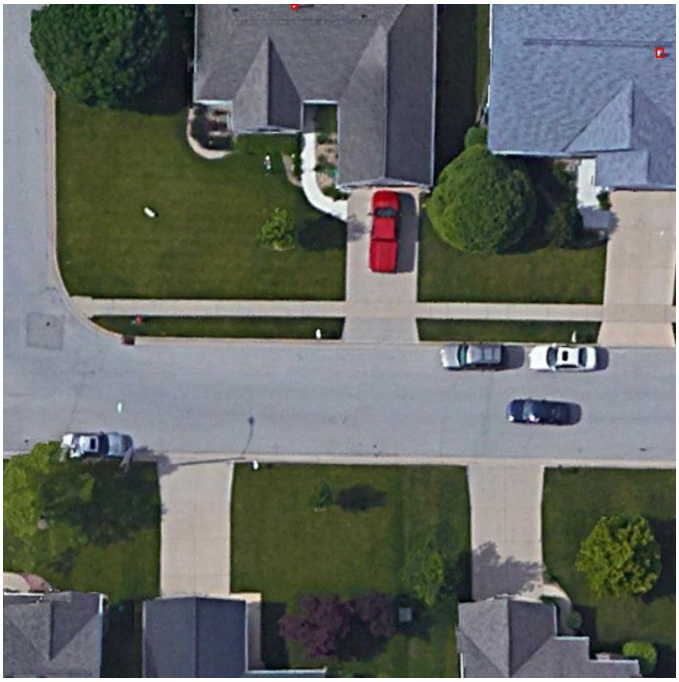


Fig. 10. Finetuned SAM3(OURS) *prompt:antenna*



Fig. 11. SAM3 Obbs without NMS *prompt:roads*



Fig. 12. SAM3 Obbs with NMS *prompt:roads*

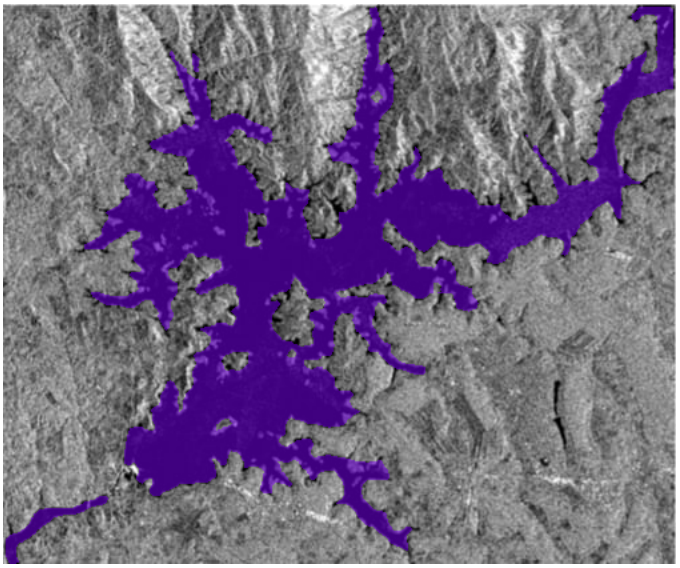


Fig. 13. Segmentation over raw IR image *prompt:water*

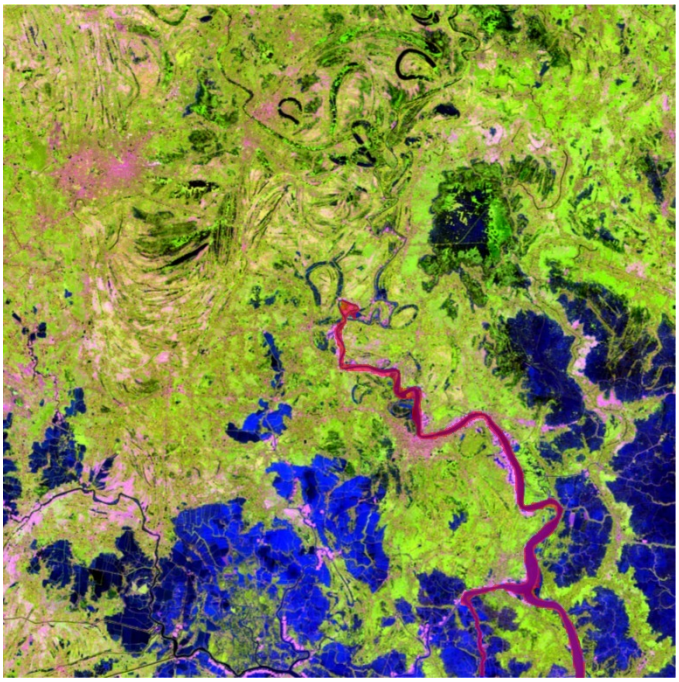


Fig. 14. Segmentation over FCC of IR *prompt:river*